

Question number	Answer	Additional guidance	Mark
6(a)	fluid mosaic (model) (1)	ACCEPT mosaic fluid IGNORE phospholipid / bilayer	(1)

Question number	Answer	Additional guidance	Mark
6(b)	A description that includes the following points: - in {a bilayer / 2 layers} (1) {phosphate / heads} pointing towards {aqueous environments / inside and_outside of the cell} (1) {fatty acids / tails} pointing {into the membrane / away from the water / towards each other} (1)	ACCEPT outwards / named environments ACCEPT inwards	(3)

Question number	Answer	Additional guidance	Mark
6(c)(i)	$r^3 = 0.027$ (1) $r = 0.3$ (1) $d = 0.6$ (1)	π on calculator $r^3 = 0.0258$ π on calculator $r = 0.2954$ π on calculator $d = 0.5908 = 0.6$ Bald answer of 0.027 = 1 mark Bald answer of 0.3 = 2 marks Bald answer of 0.6 = 3 marks	(3)

Question number	Answer	Additional guidance	Mark
6(c)(ii)	A description that includes three of the following points: • single-stranded (1) • (mRNA / mononucleotides) consist of <u>ribose</u>, phosphate and {a base / correctly named base} (1) • phosphodiester bonds between {(mono)nucleotides / phosphate and sugar} (1) • adenine, uracil, cytosine and guanine (1)	NB piece together DO NOT ACCEPT T / thymine	(3)

Question number	Answer	Additional guidance	Mark
6(c)(iii)	An explanation that includes three of the following points: • description of arrangement of {phospholipids / heads / tails} (1) • heads pointing outwards because they are polar and interact with water (environment) / hydrophobic fatty acids point away from the water (environment) (1) • mRNA surrounded by {phosphate / heads} as mRNA {is a polar molecule / has a phosphate group} (1) • {fatty acid / tails} (of both groups) facing each other because they are hydrophobic (1)	e.g. heads facing {out of the nanoparticle / mRNA} tails facing each other ACCEPT hydrophilic for polar DO NOT ACCEPT attraction of phosphate heads	(3)